

KITEWRIGHT CORE · PROBLEM STATEMENT

Kitewright Core problem statement

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/01-problem.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	Open questions answered by the decisions of KWC-DDR-001 (Amish's 2026-10-03 pre-approval); budget worded as a value-engineering target; co-design candidate named
0.1	2026-09-30	Amish Chadha	Initial scaffold

Kitewright Core problem statement

Every drone project for mountain rescue or survey ends up rebuilding the same things: autopilot setup, power, radio, a payload mount and cold-weather tuning. Commercial platforms solve this inside closed ecosystems that small teams cannot afford, inspect or adapt.

■ The problem

Demand is real. Drones have helped rescue more than 1,000 people ([DJI, 2023](#)), and heavy-lift drones now work above 5,000 m on Everest ([DJI, 2024](#)). But mountain conditions stretch every part. The standard atmosphere at 5,000 m gives 60% of sea-level air density at -17.5 °C ([Engineering ToolBox](#)), so rotors work harder while batteries deliver less. The Mavic Pro that found Rick Allen was flying far above its rated ceiling ([DroneDJ, 2018](#)).

The closed route works but locks teams in. The Matrice 350 RTK is rated to 5,000 m with standard propellers and -20 °C ([Advexure](#)), at USD 14,814 plus USD 899 per battery ([DroneFly](#)). The open route has strong parts: PX4 under BSD 3-Clause ([GitHub](#)), the Pixhawk standards including DS-014 ([Pixhawk Standards](#)), and ArduPilot for fixed wing and VTOL ([ArduPilot](#)). What is missing is an integrated, documented core, with its power bus, mount, radios and altitude tuning specified once, that every frame and payload in a family can rely on.

■ Users and context

User	Need	Context
Mountain rescue teams	A platform they can repair and adapt, with payloads that swap in the field	Search at 3,000 to 6,000 m, often in cold and wind
Glacier and hazard researchers	An inspectable platform whose logs, power and sensors are documented	Seasonal field campaigns at glacial lakes
Disaster response groups and NGOs	A lower-cost, multi-role drone that is not tied to one vendor	Floods, landslides and road cuts in mountain regions
Payload developers and students	A published mechanical, electrical and software interface to build against	University labs, makerspaces, small companies

■ Operating environment

- Operation from sea level to 5,000 m (design target), with growth to 6,000 m to be assessed.
- Ambient -20 to +45 °C (-4 to +113 °F) (target); snow, rain, dust and UV.
- Wind up to about 10 m/s in normal operation (target); gusts in mountain valleys.
- Remote sites without mains power; batteries charged from vehicles or generators.
- Radio links in steep terrain with frequent loss of line of sight.

■ Constraints

- Value-engineering target of USD 5,000 for the core, including avionics, radios, a ground station setup and spares (a hypothetical control target, not a limit; STANDARDS section 18).

- Upstream PX4 (or ArduPilot) with parameters and documented modules; no private fork of the flight stack.
- Payload mount follows Pixhawk DS-014 electrically, with a plain rail and locking pin mechanically; no bayonet.
- Power bus runs from ColdCell packs with external film heaters and a BMS thermostat; any tether input is fixed voltage with onboard DC-DC conversion.
- Same core, unchanged, in the Lift and Range frames.
- Hardware under CERN-OHL-S-2.0; software under a licence compatible with PX4 (BSD 3-Clause) or ArduPilot (GPLv3).

■ Out of scope

- Airframes (Kitewright Lift and Kitewright Range) and payloads (AvalancheScout, LakeWatch and others), which are sibling repos.
- Caged confined-space frames (deferred after the patent screen).
- Hybrid generator, fuel-cell or combustion power.
- Beyond-visual-line-of-sight approvals, satellite links and remote ID certification at this TRL.
- Aviation certification (FAA Part 107, EASA, India DGCA Drone Rules 2021) at this TRL; to be addressed if prototypes progress.
- Any weapon, targeting or military payload.

■ Prior work

Prior work	What it does	Gap for these users	Source
PX4 Autopilot	Open flight control software under BSD 3-Clause, governed by the Dronecode Foundation	Software only; no reference airframe, power bus or cold-weather integration	link
Pixhawk Payload Bus standard (DS-014)	Open standard with a 40-pin connector carrying Ethernet, USB 2.0, CAN FD, UART, trigger, PPS and battery power, plus MAVLink payload protocols	An interface standard, not a built mount or core; licence terms still to confirm	link
DJI Matrice 350 RTK	Closed enterprise drone, 55 min flight, -20 to 50 °C, 5,000 m ceiling with standard propellers, USD 14,814	Closed payload and battery ecosystem; cost beyond many mountain teams	link
DJI TB65 self-heating battery	263.2 Wh pack that heats itself below 5 °C, rated -20 to 50 °C, USD 899	Works only in DJI aircraft; used in pairs	link
ArduPilot QuadPlane	Open flight control for fixed wing aircraft with VTOL motors	Firmware only; no shared hardware core across frames	link

■ Co-design

A mountain rescue organisation or high-altitude research group working in the Himalaya or Karakoram, such as a state disaster response force unit or a university glaciology team, that can define missions, test at altitude and review the payload interface. First candidate to approach (not agreed): a university glaciology group in Jammu and Kashmir that already surveys glacial lakes in the Kashmir Himalaya, with a state disaster response force unit in the same region as the second (KWC-DDR-001, D13).

Safety: Field work with prototypes happens only with a partner that holds the local flying permissions and runs its own safety briefing; nothing here is an approval to fly.

■ Open questions answered at TRL 3

Table 1. The scaffold's open questions and the decisions that answer them (KWC-DDR-001, decided under Amish's 2026-10-03 pre-approval).

Question	Answer	Decision
Under what licence is DS-014 published, and can its connector be used freely?	Proceed with the DS-014 connector and pinout; confirm the licence before the connector is bought; if it restricts reuse, keep the same 40-pin connector with Kitewright's own open pinout	D12
PX4 or ArduPilot as the primary stack?	PX4 for the whole family; ArduPilot documented as the alternative	D1
Which radio bands and power levels at target sites?	900 MHz class radios set per country; for India the 865 to 867 MHz licence-exempt band is the first candidate, to confirm with the partner	D10
What payload mass and power classes?	One class for version 1: 5 kg and 100 W	D4
Remote ID and flight logging?	Every flight logged on the controller; a remote ID module goes on the CAN bus when a country's rules require it	D14

■ Safety

Safety: The problem involves lithium packs, spinning propellers and loads carried over people and terrain. Every Kitewright document that describes these keeps a safety section; the core's precis (KWC-PRC-001) and build plan (KWC-BLD-001) set out the hazards and the safety stops. Civilian use only; no weapons, targeting or military payloads.